10 Jump the Shark

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For this assignment, you'll install and run Wireshark. Then you'll capture network data and filter it in various ways.

Setup

Install Wireshark in the location of your choice. If you want a safe playground, I suggest using a Windows 10 virtual machine from vcm.duke.edu. If you use your personal laptop instead, just be aware that you will be submitting information about your network configuration and traffic. If that is a concern, use a VCM machine instead.

Wireshark is here:

https://www.wireshark.org/download.html

Capture

Run Wireshark and capture the following traffic.

1. Run `ping` (ICMP) to or from the machine where you are running Wireshark.
2. Retrieve a web page (HTTP) using a browser or `curl`. Use a domain name that you haven't seen in a while, to force your computer to do an explicit DNS lookup.
3. Retrieve a web page (HTTPS) using a browser or `curl`.

You can do these all in one capture, or you can do separate captures if it's easier.

Don't forget to stop the capture when you have everything you want.

Submit

Answer the following questions and submit the following information.

1. What is the MAC address and IP address on the computer where you have installed Wireshark? Show the output of `ipconfig /all`(Windows), `ifconfig` (Linux), or a screen shot if you locate this information in a GUI somewhere. (I think `ifconfig` works on MacOS too.)
2. Filter your capture to show the pings only. Export these packets as a separate file and submit the file.
3. Create an exported file that tells the story of the HTTP web page retrieval. Your exported file should include the DNS lookup, the TCP setup packets, the HTTP request, and the HTTP response from the server. You might want to use the packet marking feature (Ctrl-M) to designate which packets you will export.
4. Follow the stream of your HTTP request (right-click, then **Follow** > **HTTP Stream**). Submit a screen shot, or copy and paste the contents.
5. Export the packets that show the HTTPS connection, including the TCP setup and signaling. Hint: you can follow the TCP stream for this one.

In summary, you will be submitting three packet capture files (#2, #3, and #5) and answering #1 and #4 with copy-pasted output or screen shots.